Orchestrator

EventQueue

ConstellationAgent (Editor)

System running normally

enqueue(TASK_COMPLETED(A)) (t0)

1

acquire(assign_lock) (t1)

2

[LOCKED] suspend new assignments

start edit based on A (t2)

3

TASK_COMPLETED(B) arrives (t3, queued)

4

TASK_COMPLETED(C) arrives (t4, queued)

5

ΔA (edit result ready)

6

apply ΔA, validate DAG, synchronize runtime state (t5)

7

publish CONSTELLATION_MODIFIED(ΔA)

8

DAG updated after ΔA

start edit based on [B, C] (t6)

9

Δ[B,C] (edit result ready)

10

apply Δ[B,C], validate DAG, synchronize runtime state (t7)

11

publish CONSTELLATION_MODIFIED(Δ[B,C])

12

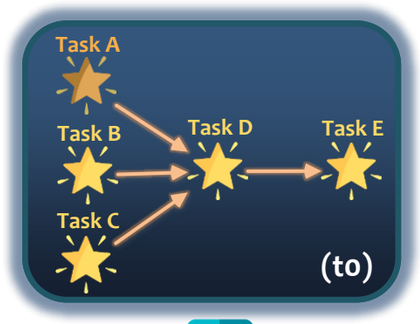
DAG updated after Δ[B,C]

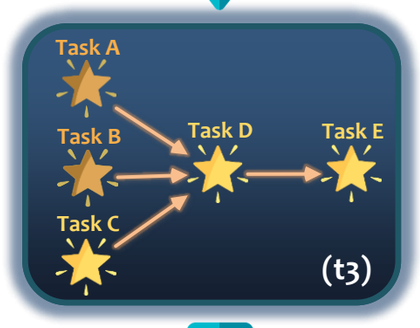
All edits applied and synchronized

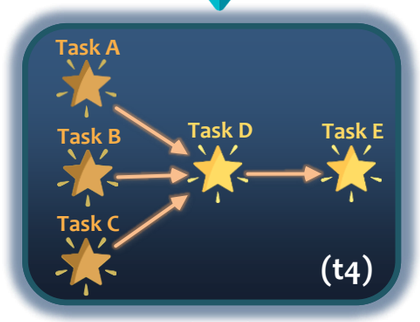
release(assign_lock) (t8)

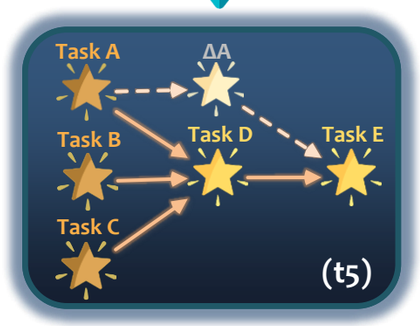
13

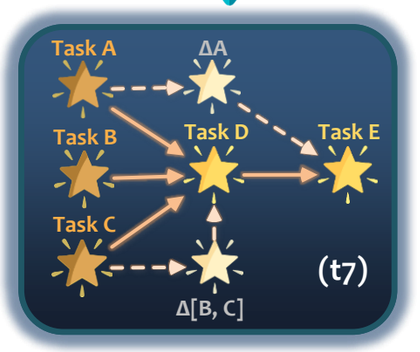
[UNLOCKED] resume scheduling











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